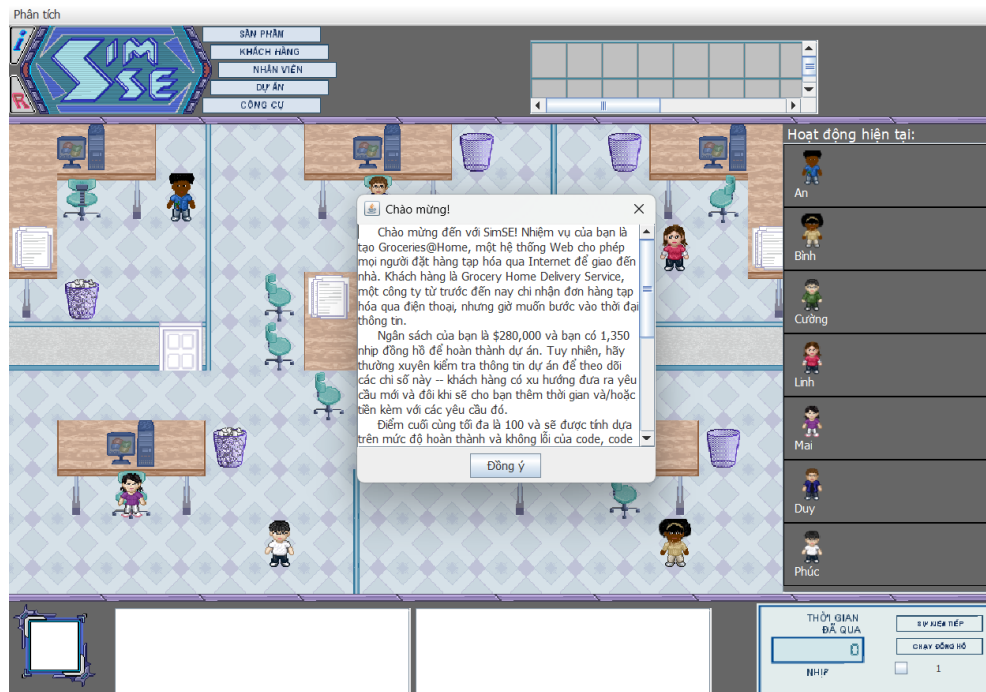


Software Engineering - Lab 8: Agile Development & Project Management

Overview

In this assignment, you will apply the intended workflow of **Waterfall** and **XP process models** in a simulated project. Below is a sample screenshot of the simulation.



Learning outcomes:

- Run the **SimSE-VN** simulations locally and identify the major UI regions used to manage the project.
- Apply the intended workflow of Waterfall and XP process models in a simulated project.
- Inspect employee skills, artifacts, project status, tools, and customer information before assigning work.
- Advance simulation time deliberately, observe effects, and revise decisions based on evidence.
- Compare how sequencing, feedback, testing, and customer interaction differ across the two assigned models.

Acknowledgment and Source Attribution

This assignment uses SimSE, a software engineering process simulation originally created by Emily Oh Navarro and Andre van der Hoek at the University of California, Irvine. The original SimSE project and its educational simulation concepts should be credited to those creators.

The SimSE-VN package used in this course is a Vietnamese localized version prepared for instructional use. The localization, build packaging, and lab instructions support this course activity and do not replace the original SimSE authorship. Original SimSE site:

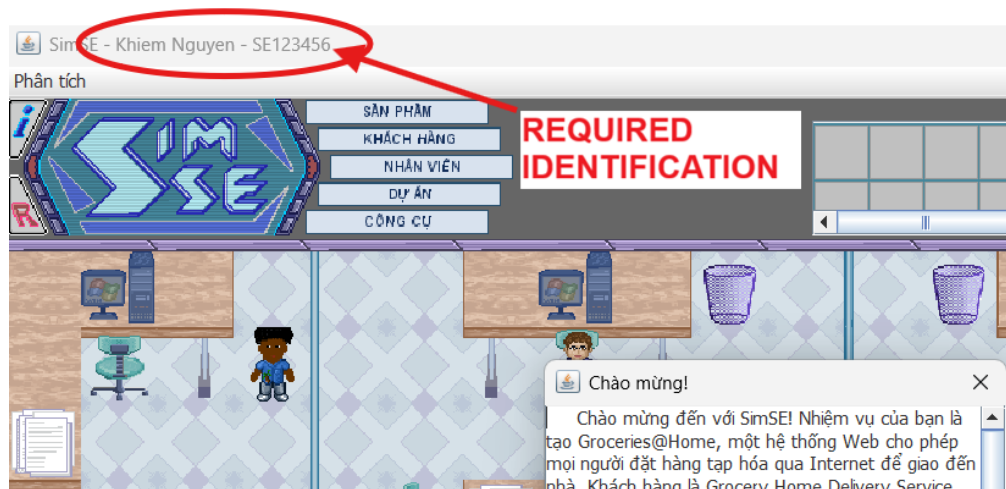
<http://www.ics.uci.edu/~emilyo/SimSE>.

If you refer to the tool in your report, cite it as: SimSE, by Emily Oh Navarro and Andre van der Hoek, University of California, Irvine.

Build and Run

Required simulation rebuild

Before collecting evidence screenshots, rebuild once with your own **name** and **student ID**. This does not change the simulation rules; it only changes the window title used for submission tracking. After the rebuild, run Waterfall and XP from bin/simulations as usual. Your screenshot title bar must read like **"SimSE - Nguyen Van A - SE123456"**, as shown in the below.



Use **your real name** and **real student ID**; do not submit screenshots that still show a classmate name, the instructor sample, or the default SimSE title.

Platform	Command
Windows	<pre>cd simse-vn .\tools\build-simulations-windows.ps1 -StudentName "Nguyen Van A" -StudentId "SE123456"</pre>
macOS	<pre>cd simse-vn chmod +x ./tools/build-simulations-macos.sh ./tools/build-simulations-macos.sh -StudentName "Nguyen Van A" -StudentId "SE123456"</pre>

Use the ready-to-run builds

The required runnable folders for this assignment are:

Model	Folder	Main JAR
Waterfall	simse-vn\bin\simulations\waterfall	simse-waterfall-vn.jar
XP	simse-vn\bin\simulations\xp	simse-xp-vn.jar

Submission Guidelines

1. **Format:** Submit your assignment as **two image files**.
2. **Naming Convention:** lab8_<taskA/B/C>_<fullname>_<studentID>

Where:

- <taskA/B/C> is the identification of the task you submit.
- <fullname> is your full name.
- <studentID> is your student ID.

Required Tasks

Task A - Waterfall

Goal

Complete the project by following a mostly sequential lifecycle. The main challenge is to avoid rushing into coding before requirements and design artifacts are mature enough.

Start Waterfall simulation at:

simse-vn\bin\simulations\waterfall\simse-waterfall-vn.jar

*A simulation **score of over 80** gets the full score for this part in the rubric.*

Suggested workflow

1. Inspect the customer, project, requirement artifact, design artifact, code artifact, and team members.
2. Assign requirements work first. Review and correct requirements before moving to design.
3. Assign design work after requirements have enough quality. Review and correct design defects.
4. Assign coding only after the design is ready. Use stronger coding experience where possible.
5. Inspect and correct code before integration or system-level test work.
6. Use testing actions near the end. Avoid treating final testing as a substitute for earlier review.
7. Deliver only when project and artifact evidence suggests the product is ready.

Guidance

Feel free to look at [official source](#) for more details. Below are other people playthroughs that you can refer to:

- [Video 1](#)
- [Video 2](#)

Task B - Extreme Programming (XP)

Goal

Use short feedback cycles, customer involvement, testing, and frequent small releases. The main challenge is keeping enough discipline while moving quickly.

Start Extreme Programming (XP) simulation at:
`simse-vn\bin\simulations\xp\simse-xp-vn.jar`

*A simulation **score of over 80** gets the full score for this part in the rubric.*

Suggested workflow

1. Inspect employee skills and customer/project objects before assigning the first task.
2. Choose a small unit of work rather than trying to complete everything at once.
3. Use XP-style activities such as customer discussion, pair programming, testing, integration, refactoring, or release when they are available in the action menu.
4. Advance a few ticks, inspect feedback, and adjust the next small task.
5. Use tests and corrections repeatedly. Do not leave testing only for the final moments.
6. Compare your XP run against the more plan-driven models in your final discussion.

Guidance

Feel free to look at [official source](#) for more details. Below are other people playthroughs that you can refer to:

- [Video 1](#)
- [Video 2](#)

(Optional) Task C- Incremental

Complete a screenshot proof for completion of the Incremental simulation for **10-point-bonus**.

Grading Rubrics

Part	Criteria	Points
Task A - Waterfall	Submit a correct screenshot proof	10
	Waterfall score: $\min(\text{final Waterfall score}, 80) / 80 \times 40$.	40
Subtotal		50
Task B - Extreme Programming (XP)	Submit a correct screenshot proof	10
	Waterfall score: $\min(\text{final XPs score}, 80) / 80 \times 40$.	40
Subtotal		50
TOTAL		100